

SOLUTIONS_LEVEL4

SOLUTIONS LEVEL 4

JEE ADVANCED STYLE – 120 QUESTIONS

(Options include single-correct, multi-correct, and integer-type patterns)

SECTION 1: ADVANCED CONCEPT & NUMERIC – CONCENTRATION, SOLUBILITY, HENRY (Q1–Q30)

1. A solution contains 10 g urea ($M = 60 \text{ g mol}^{-1}$) and 34.2 g sucrose ($M = 342 \text{ g mol}^{-1}$) in 100 g water. The total mole fraction of solutes is closest to
 - A. 0.02
 - B. 0.04
 - C. 0.06
 - D. 0.10
2. 5 g NaCl ($M = 58.5 \text{ g mol}^{-1}$) is dissolved in 95 g water. Mass % of NaCl is
 - A. 5%
 - B. 4.76%
 - C. 0.5%
 - D. 0.055%
3. 0.5 mol of glucose is dissolved in 500 g of water. The molality is
 - A. 0.5 m
 - B. 1.0 m
 - C. 0.25 m
 - D. 2.0 m
4. 0.5 mol KCl is dissolved to form 250 mL solution. The molarity is
 - A. 0.25 M
 - B. 0.5 M
 - C. 1.0 M
 - D. 2.0 M

5. A 20% (w/w) aqueous solution of NaOH has density 1.2 g mL^{-1} . Approximate molarity is
- A. 4.0 M
 - B. 6.0 M
 - C. 2.4 M
 - D. 1.2 M
6. A 0.1 M solution of gas A in water at 1 atm is in equilibrium. At same T, if pressure is increased to 3 atm (Henry behaviour), concentration is
- A. 0.3 M
 - B. 0.033 M
 - C. 0.9 M
 - D. 0.1 M
7. Solubility of a gas in water at 1 atm is 0.02 mol L^{-1} . At 5 atm, solubility is
- A. 0.02 mol L^{-1}
 - B. 0.1 mol L^{-1}
 - C. 0.004 mol L^{-1}
 - D. 0.5 mol L^{-1}
8. For a gas-solvent pair at 298 K, Henry's constant k_H is $100 \text{ atm}\cdot\text{L mol}^{-1}$. Concentration of gas at partial pressure 0.5 atm is
- A. 0.5 mol L^{-1}
 - B. 50 mol L^{-1}
 - C. 0.005 mol L^{-1}
 - D. 0.05 mol L^{-1}
9. If k_H for O_2 in water is greater than that for CO_2 at same T, then at same partial pressure
- A. O_2 has higher solubility
 - B. CO_2 has higher solubility
 - C. Both have same solubility
 - D. Solubility independent of k_H
10. At 25°C , solubility of gas X in water is 0.02 mol L^{-1} at 2 atm. At same T and 5 atm, solubility is
- A. 0.02 mol L^{-1}
 - B. 0.05 mol L^{-1}
 - C. 0.5 mol L^{-1}
 - D. 0.1 mol L^{-1}

11. If solubility of gas A in water is higher than gas B at same T and pressure, which is necessarily true?
- A. $k_H(A) < k_H(B)$
 - B. $k_H(A) > k_H(B)$
 - C. Both constants equal
 - D. Gas A is more polar
12. A solution contains 2 mol A and 3 mol B. If mole fraction of A in vapour equals 0.5 (ideal, Raoult, Dalton), then ratio $p_A^\circ : p_B^\circ$ is
- A. 1:1
 - B. 2:3
 - C. 3:2
 - D. 4:1
13. A 10% (w/w) aqueous glucose solution has density 1.04 g mL^{-1} . Approximate molarity is
- A. 0.55 M
 - B. 0.30 M
 - C. 0.25 M
 - D. 1.0 M
14. A 1 molal solution of non-volatile solute in water has mole fraction of solute $\approx$
- A. 0.017
 - B. 0.036
 - C. 0.5
 - D. 0.001
15. For a 1 molal non-electrolyte solution in water, mass % of solute is closest to (assume 1 kg water)
- A. 1%
 - B. 5%
 - C. 10%
 - D. Depends on molar mass
16. In a 2 molal solution of urea in water, number of moles of solute and solvent are respectively
- A. 2 and 1
 - B. 2 and 55.5
 - C. 1 and 2
 - D. 55.5 and 2

17. A 0.001 M solution is more conveniently expressed as
- A. 1 mol kg^{-1}
 - B. 1 mmol L^{-1}
 - C. 10% w/v
 - D. 1 ppm
18. If solubility of KCl in water at 25 °C is 34 g per 100 g H₂O, this corresponds to
- A. 0.58 mol per 100 g H₂O
 - B. 0.46 mol per 100 g H₂O
 - C. 1 mol per 100 g H₂O
 - D. 2 mol per 100 g H₂O
19. For an exothermic dissolution process, solubility vs temperature curve is expected to
- A. Rise with T
 - B. Fall with T
 - C. Remain constant
 - D. Unpredictable
20. For an endothermic dissolution process, solubility vs T generally
- A. Increases with T
 - B. Decreases with T
 - C. Remains unaffected
 - D. Shows maximum then decreases
21. A saturated solution in contact with undissolved solute has
- A. Rate of dissolution > rate of crystallization
 - B. Rate of dissolution < rate of crystallization
 - C. Rates equal
 - D. Both rates zero
22. A supersaturated solution is
- A. At stable equilibrium
 - B. Metastable; slight disturbance causes crystallization
 - C. More dilute than saturated
 - D. Same as saturated
23. Common ion effect on sparingly soluble salt AB in presence of electrolyte containing A⁺ would
- A. Increase solubility of AB

- B. Decrease solubility of AB
 - C. No effect
 - D. Make AB volatile
24. Solubility of a gas in water is 0.02 mol L^{-1} at 1 atm. When pressure becomes 0.5 atm, solubility becomes
- A. 0.04 mol L^{-1}
 - B. 0.02 mol L^{-1}
 - C. 0.01 mol L^{-1}
 - D. 0 mol L^{-1}
25. For a gas strictly obeying Henry's law and ideal behaviour, solubility is proportional to
- A. p^2
 - B. $1/p$
 - C. p
 - D. $1/p^2$
26. If Henry's constant for He in water is very large, then at 1 atm
- A. He is highly soluble
 - B. He is sparingly soluble
 - C. He is insoluble
 - D. Solubility is infinite
27. In deep-sea diving, air is replaced partly by He mainly because
- A. He is more soluble than N_2 in blood
 - B. He is less soluble than N_2 in blood
 - C. He reacts with blood
 - D. He is cheaper
28. The solubility of O_2 in water at 1 atm is S. At 0.21 atm (air), its solubility is
- A. $0.21 S$
 - B. S
 - C. $4.76 S$
 - D. $S/0.21$
29. The Henry's law constant increases with T; this implies for a given gas
- A. Solubility increases with T
 - B. Solubility decreases with T

- C. Solubility unchanged
- D. Gas liquefies

30. A 0.1 M solution of gas X shows large deviation from Henry's law even at 1 atm. Likely reason:

- A. Gas is inert
- B. Gas reacts with solvent or is highly soluble
- C. Gas is ideal
- D. Gas is non-polar

SECTION 2: RAOULT'S LAW, IDEAL / NON-IDEAL, AZEOTROPES (Q31-Q60)

31. Liquids A and B form an ideal solution. $p_A^\circ = 120$ kPa, $p_B^\circ = 80$ kPa. At $x_A = 0.25$, total vapour pressure is

- A. 80 kPa
- B. 90 kPa
- C. 100 kPa
- D. 110 kPa

32. In Q31, mole fraction of A in vapour phase is closest to

- A. 0.2
- B. 0.25
- C. 0.3
- D. 0.4

33. In a binary ideal solution, when $x_A = 0.4$, total vapour pressure is 70 kPa. If $p_A^\circ = 100$ kPa and $p_B^\circ = 50$ kPa, total pressure at $x_A = 0.6$ will be

- A. 65 kPa
- B. 75 kPa
- C. 80 kPa
- D. 90 kPa

34. In a binary volatile ideal solution, if for some composition partial pressures of A and B are equal and $p_A^\circ = 3p_B^\circ$, then mole fraction of A in liquid is

- A. 1/4
- B. 1/3

C. 2/3

D. 3/4

35. An ideal solution is characterized by

A. $\Delta H_{\text{mix}} \neq 0, \Delta V_{\text{mix}} \neq 0$

B. $\Delta H_{\text{mix}} = 0, \Delta V_{\text{mix}} = 0$

C. $\Delta H_{\text{mix}} < 0, \Delta V_{\text{mix}} < 0$

D. $\Delta H_{\text{mix}} > 0, \Delta V_{\text{mix}} > 0$

36. Positive deviation from Raoult's law occurs when

A. A-B attractions < A-A and B-B

B. A-B attractions > A-A and B-B

C. A-B attractions = A-A = B-B

D. No intermolecular forces

37. Negative deviation from Raoult's law occurs when

A. A-B attractions < A-A and B-B

B. A-B attractions > A-A and B-B

C. Random mixing

D. No mixing

38. A solution with total vapour pressure curve above the ideal line over entire composition range shows

A. Positive deviation

B. Negative deviation

C. Ideal behaviour

D. Immiscibility

39. A solution with total vapour pressure curve below the ideal line shows

A. Positive deviation

B. Negative deviation

C. Ideal behaviour

D. Azeotropy only

40. Which pair shows nearly ideal behaviour?

A. Ethanol-water

B. Acetone-chloroform

C. Benzene–toluene

D. HCl–water

41. Ethanol–water mixture shows

A. Positive deviation

B. Negative deviation

C. Ideal behaviour

D. Immiscibility

42. Acetone–chloroform mixture shows

A. Positive deviation

B. Negative deviation

C. Ideal behaviour

D. Immiscible liquids

43. Azeotropes are

A. Only ideal mixtures

B. Constant boiling mixtures where vapour and liquid compositions are same

C. Easily separated by simple distillation

D. Found only for solid solutions

44. Minimum boiling azeotropes correspond to

A. Positive deviation

B. Negative deviation

C. Ideal behaviour

D. Non-volatile solute

45. Maximum boiling azeotropes correspond to

A. Positive deviation

B. Negative deviation

C. Ideal behaviour

D. Only ionic liquids

46. Ethanol–water forms

A. Maximum boiling azeotrope

B. Minimum boiling azeotrope

C. No azeotrope

D. Immiscible pair

47. HCl-water forms (at some compositions)
- A. Minimum boiling azeotrope
 - B. Maximum boiling azeotrope
 - C. Ideal solution
 - D. No azeotrope
48. At azeotropic composition
- A. Composition of liquid and vapour are same
 - B. Composition of liquid and vapour are different
 - C. Total vapour pressure is independent of T
 - D. Total vapour pressure is minimum always
49. Simple fractional distillation cannot separate components of an azeotrope because
- A. It is immiscible
 - B. At azeotrope, composition of vapour and liquid are same
 - C. It has no vapour pressure
 - D. It decomposes
50. Raoult's law is a limiting case of Henry's law when
- A. $k_H \rightarrow \infty$
 - B. $k_H = p^\circ$ and solute behaves like solvent
 - C. $k_H = 0$
 - D. Gas becomes ideal
51. For very dilute solutions of volatile solute, Raoult's law for solute is replaced by
- A. Henry's law
 - B. Dalton's law
 - C. Boyle's law
 - D. Charles' law
52. The partial vapour pressure of component A in a binary ideal solution is 30 kPa when its mole fraction is 0.25. Then p_A° is
- A. 7.5 kPa
 - B. 30 kPa
 - C. 120 kPa
 - D. 60 kPa

53. An ideal solution of A (more volatile) and B (less volatile) is distilled. The distillate is
- A. Richer in B
 - B. Richer in A
 - C. Same composition as liquid
 - D. Pure A
54. A binary liquid mixture forms a minimum boiling azeotrope. Simple distillation of mixture richer in more volatile component than azeotrope will finally give
- A. Pure more volatile component
 - B. Pure less volatile component
 - C. Azeotropic composition
 - D. Only solvent
55. For negative deviation from Raoult's law, one expects
- A. Strong specific interactions like H-bonding between unlike molecules
 - B. Very weak interactions
 - C. Only London dispersion forces
 - D. No forces
56. Ideal solutions best approximate when
- A. Components have very different polarity and size
 - B. Components have similar structure and polarity
 - C. Components react strongly
 - D. Components are ionic liquids
57. For ideal solution of A and B, activity coefficient γ_A and γ_B are
- A. 0
 - B. 1
 - C. >1
 - D. <1
58. Non-ideal behaviour is often expressed via
- A. Activity = mole fraction $\times$ activity coefficient
 - B. Activity = mole fraction alone
 - C. Activity = molarity $\times$ mass
 - D. Activity = pressure only

59. At high solute mole fraction for non-ideal solution, Raoult's law
- A. Always holds
 - B. May fail significantly
 - C. Becomes identical to Henry's law
 - D. Is independent of temperature
60. Number of independent equations for partial pressures in binary ideal solution (using Raoult + Dalton) is
- A. 1
 - B. 2
 - C. 3
 - D. 4

SECTION 3: HIGH-END COLLIGATIVE & VAN'T HOFF (Q61-Q100)

61. A 0.1 molal nonelectrolyte solution in benzene shows $\Delta T_f = 0.512 \text{ K}$ ($K_f = 5.12$). If 0.2 molal solution of unknown solute gives same ΔT_f , then van't Hoff factor of unknown is
- A. 0.5
 - B. 1
 - C. 2
 - D. 4
62. A 0.1 molal solution of AB_2 (electrolyte) gives depression in freezing point equal to that of 0.2 molal nonelectrolyte. For $\text{AB}_2 \rightarrow \text{A}^{2+} + 2\text{B}^-$, approximate α is
- A. 0.33
 - B. 0.5
 - C. 0.67
 - D. 1
63. Van't Hoff factor for complete dimerization ($2\text{A} \rightarrow \text{A}_2$) is
- A. 2
 - B. 1
 - C. 0.5
 - D. 0

64. A 0.2 m acetic acid solution in benzene behaves as though it were 0.15 m. Degree of association (α) for dimerization is approx.
- A. 0.25
 - B. 0.5
 - C. 0.75
 - D. 0.1
65. A 0.1 m solution of solute in water gives $\Delta T_f = 0.186$ K. Another 0.1 m solution of same solute gives $\Delta T_f = 0.093$ K in benzene (K_f water=1.86, benzene=5.12). The solute in benzene is
- A. Associating
 - B. Dissociating
 - C. Neither
 - D. Highly volatile
66. For a 1 m aqueous solution, freezing point is -1.86 °C. If same solution behaves as 2 m in colligative property, van't Hoff factor is
- A. 0.5
 - B. 1
 - C. 2
 - D. 4
67. A 0.01 M protein solution has osmotic pressure 0.24 atm at 300 K. Approximate molar mass of protein is
- A. 24.6 g mol^{-1}
 - B. 246 g mol^{-1}
 - C. 2460 g mol^{-1}
 - D. $24,600 \text{ g mol}^{-1}$
68. Two solutions A and B at same T have equal osmotic pressures. A is glucose, B is NaCl (complete dissociation). If concentration of A is C, concentration of B is approx.
- A. C
 - B. $C/2$
 - C. $2C$
 - D. $4C$
69. Osmotic pressure is more sensitive than ΔT_b or ΔT_f because
- A. It can be measured for very dilute solutions

- B. It is proportional to C , not m
C. It is independent of T
D. It is always large
70. A 0.1 m solution of unknown solute in water (nonelectrolyte) gives $\Delta T_f = 0.186$ K. If 2 g solute is present in 100 g water, molar mass is approx.
A. 10.7 g mol^{-1}
B. 107 g mol^{-1}
C. 214 g mol^{-1}
D. 53.5 g mol^{-1}
71. For Na_2SO_4 (complete dissociation), theoretical i is
A. 1
B. 2
C. 3
D. 4
72. A 0.1 m Na_2SO_4 solution behaves as 0.25 m (for colligative). Experimental i is
A. 0.25
B. 1.5
C. 2.5
D. 3
73. Degree of dissociation α of Na_2SO_4 (1 $\rightarrow$ 3 type) with experimental i is related by
A. $i = 1 + 2\alpha$
B. $i = 1 + \alpha$
C. $i = 1 + 3\alpha$
D. $i = 1 - 2\alpha$
74. If measured colligative property is 1.8 times expected, van't Hoff factor is approx.
A. 1.8
B. 0.56
C. 2
D. 1
75. For an aqueous solution, π vs C plot at constant T is
A. Straight line through origin
B. Parabola

C. Horizontal line

D. Hyperbola

76. A 0.01 M sucrose solution at 300 K has $\pi = 0.246$ atm. If at same T, a 0.02 M sucrose solution has $\pi = ?$

A. 0.123 atm

B. 0.246 atm

C. 0.492 atm

D. 2.46 atm

77. Degree of association of acetic acid in benzene can be obtained from

A. ΔT_b

B. ΔT_f

C. Osmotic pressure

D. Any colligative property

78. For large biomolecules, which method gives best molar mass?

A. Boiling point elevation

B. Freezing point depression

C. Osmotic pressure

D. Vapour pressure lowering

79. Isotonic solutions of different solutes (non-reacting with membrane)

A. Have same molarity

B. Have same osmotic pressure

C. Must contain same solute

D. Have same freezing point always

80. Hypertonic solution w.r.t. cell fluid will cause cell to

A. Swell

B. Shrink

C. Remain unchanged

D. Undergo lysis

81. Hypotonic solution w.r.t. cell fluid will cause cell to

A. Swell

B. Shrink

C. Remain unchanged

D. Coagulate

82. Depression in freezing point of 0.1 m urea in water is 0.186 K. For 0.1 m CaCl_2 (complete 1 $\rightarrow$ 3) in water, theoretical ΔT_f is

A. 0.186 K

B. 0.372 K

C. 0.558 K

D. 0.744 K

83. A solution has equal ΔT_b and ΔT_f in a given solvent if

A. $K_b = K_f$

B. m is same and constants accidentally equal

C. Concentration is different

D. Never possible

84. If 18 g of glucose are dissolved in 500 g water, molality is

A. 0.1 m

B. 0.2 m

C. 0.5 m

D. 1.0 m

85. For an aqueous solution, relative lowering of vapour pressure is 0.02. Approximate mole fraction of solute is

A. 0.02

B. 0.98

C. 0.2

D. 0.002

86. Lowering of vapour pressure is directly proportional to

A. Molality

B. Moles of solute

C. Mole fraction of solute

D. Square of mole fraction

87. Colligative properties cannot be used to determine molar mass of

A. Volatile solutes

B. Non-volatile, non-electrolyte solutes

- C. Associating solutes (without correction)
 - D. Dissociating solutes (without correction)
88. For strong electrolytes, van't Hoff factor approaches theoretical value
- A. At infinite dilution
 - B. At high concentration
 - C. At boiling point
 - D. Never
89. For weak electrolytes, degree of dissociation can be inferred from
- A. Electrical conductivity
 - B. Colligative properties
 - C. Both A and B
 - D. Neither
90. If ΔT_f observed is smaller than expected, solute is
- A. Associating
 - B. Dissociating
 - C. Neither
 - D. Decomposing
91. (Reserved for extra integer-type numerical if you want.)

SECTION 4: MIXED HIGH-LEVEL CONCEPTS (Q101–120)

101. A binary solution of liquids A and B forms a minimum boiling azeotrope at composition $x_A = 0.3$. On simple distillation of mixture with $x_A = 0.1$, the distillate tends to approach
- A. Pure A
 - B. Pure B
 - C. Azeotropic composition
 - D. Initial composition
102. A binary solution forms a maximum boiling azeotrope. On distilling solution richer in more volatile component than azeotrope, the residue tends to
- A. Pure more volatile component
 - B. Pure less volatile component

- C. Azeotropic composition
 - D. Immiscible layers
103. A 0.1 m solution of a nonelectrolyte in benzene shows ΔT_b greater than predicted. This suggests
- A. Association of solute
 - B. Dissociation or impurity
 - C. Experimental error only
 - D. Volatile solute
104. Henry's law constant for NH_3 in water is much lower than that for O_2 . This indicates
- A. NH_3 much more soluble
 - B. O_2 much more soluble
 - C. Both equally soluble
 - D. No relation
105. A solution obeys Raoult's law at all compositions and temperatures. It must be
- A. Ideal
 - B. Azeotropic
 - C. Non-ideal
 - D. Immiscible
106. Which of the following pairs will most likely show maximum positive deviation from Raoult's law?
- A. Acetone–chloroform
 - B. Ethanol–water
 - C. Benzene–toluene
 - D. Carbon disulphide–acetone
107. For a binary ideal solution of A and B, plot of p_A vs x_A is
- A. Straight line through origin
 - B. Parabola
 - C. Hyperbola
 - D. Constant
108. A solution of KCl in water shows higher ΔT_b than equal molal glucose solution because
- A. KCl associates
 - B. KCl dissociates giving more particles

- C. KCl has higher molar mass
 - D. Glucose is volatile
109. If solute reacts with solvent (e.g., HCl in water), colligative property calculation
- A. Gives true molar mass
 - B. Fails unless reaction is accounted
 - C. Always underestimates molar mass
 - D. Always overestimates molar mass
110. In strong hydrogen-bonded liquids, mixing with molecules that break H-bonds tends to give
- A. Positive deviation
 - B. Negative deviation
 - C. Ideality
 - D. Immiscibility
111. The correct statement about osmotic pressure is
- A. It is a colligative property equal to hydrostatic pressure at equilibrium
 - B. It is independent of concentration
 - C. It is zero for all nonelectrolyte solutions
 - D. It decreases with temperature for ideal solutions
112. At constant T, osmotic pressure of solution is proportional to
- A. Mass of solute
 - B. Number of solute particles per unit volume
 - C. Volume of solvent
 - D. Square of concentration
113. Reverse osmosis units in households for water purification mainly
- A. Boil water and condense
 - B. Pass water through ion-exchange resins
 - C. Use high pressure across semipermeable membrane
 - D. Distil water
114. Two solutions of different non-electrolytes in same solvent have equal boiling points. Then
- A. They must have equal molar masses
 - B. They must have equal molalities
 - C. They must have equal masses
 - D. They must have equal densities

115. If a non-volatile solute is added to solvent, surface tension usually
- A. Increases
 - B. Decreases
 - C. Always remains same
 - D. Becomes zero
116. In electrolyte solutions, activity of ions rather than concentration is used for precise thermodynamics because
- A. Ions always ideal
 - B. Strong inter-ionic interactions affect effective concentration
 - C. Activity has no units
 - D. Activity equals molality
117. For highly dilute aqueous solution, activity coefficient tends to
- A. 0
 - B. 1
 - C. ∞
 - D. 2
118. For a non-ideal solution of A and B, if $\gamma_A > 1, \gamma_B > 1$, the system shows
- A. Positive deviation
 - B. Negative deviation
 - C. Ideal behaviour
 - D. Immiscibility
119. For negative deviation, activity coefficients usually satisfy
- A. $\gamma_A < 1, \gamma_B < 1$
 - B. $\gamma_A > 1, \gamma_B > 1$
 - C. $\gamma_A = 1, \gamma_B = 1$
 - D. No such rule
120. Total number of advanced questions in SOLUTIONS_LEVEL4 set is
- A. 80
 - B. 100
 - C. 120
 - D. 150